TECHNICAL DESIGN DOCUMENT (TDD)

SarathiPOD: AI-Powered Proof of Delivery & Accelerated Billing Platform

GitHub: https://github.com/SaiTejesh09/sarathipod | **Live App:** https://sarathipod-production.up.railway.app
Client: Company A Logistics (Pvt Ltd), MIDC Waluj, Chhatrapati Sambhajinagar, Maharashtra | **Version:** 1.0.0

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1. Executive Summary & Problem Formulation

Company A is a full-truckload (FTL) transporter operating from Chhatrapati Sambhajinagar, Maharashtra, moving cement, steel, and packaged foods across long-haul inter-state corridors spanning 800+ km (Nagpur, Pune, Indore, Raipur, Hyderabad). Fleet: 140 own trucks + ~200 hired market trucks; 210 staff; annual revenue ■75–90 Crore (modeled at ■84.48 Cr/year across 2,200 trips/month at ■32,000 average freight).

Quantified Business ROI & Annual Value (Calculated directly from Brief):
• **Trapped Working Capital:** Cutting POD-to-invoice from 18 days to 4 days brings billing 14 days forward: **■84.48 Cr × 14 / 365 = ■3.24 Crore** of working capital unlocked.
• **Interest Savings:** At 12% p.a. bank borrowing rate, saves **■38.9 Lakh / year** in interest.
• **Disputed Freight Saved:** 1.5% of trips (396 trips/yr) lose PODs, causing 50% freight deduction (396 × ■16,000 = **■63.36 Lakh / year**).
• **Total Annual Value:** ■38.9 Lakh + ■63.36 Lakh = **■1.02 Crore / year**.
• **Budget & Payback:** Project budget ■7.5 Lakh pays back in **27 Days (under 1 month!)**.

Assumptions Register (Labeled as per submission rules):

• **[ASSUMPTION-01: Dead Zones]** Highway corridors have dead zones; capture operates offline in IndexedDB with client-side canvas compression <300 KB.
• **[ASSUMPTION-02: Low-Literacy Drivers]** UI features 1 big button, no typing, and voice guidance in Marathi (mr-IN) and Hindi (hi-IN).
• **[ASSUMPTION-03: Hired Drivers]** 200 hired market drivers will not install an app; zero-install SMS upload link (/p/:token) provided.
• **[ASSUMPTION-04: DPDP Act 2023]** Consignee names, signatures, and driver phone numbers are masked in audit logs and client views.
• **[ASSUMPTION-05: Tally Accounting]** Billing requires Tally Prime Sales Voucher XML with User Defined Fields (UDF) for LR and Vehicle.

2. Architecture & Technology Rationale

• **Frontend:** React 18 + Vite + Tailwind CSS. Fast, responsive, mobile-first design, gzip 66 KB bundle.
• **Regional Audio:** Native Web Speech Synthesis with Marathi and Hindi audio guidance and pleasant audio chimes.
• **Edge Compression:** HTML5 Canvas API resizes camera photos iteratively until file size is strictly under 300 KB.
• **Offline Storage:** IndexedDB queue captures PODs offline and flushes automatically when 4G resumes.
• **Backend:** Python 3.11 + FastAPI with asynchronous request handling and Pydantic validation.
• **Document Processing:** Pillow for stamp ink clustering, signature edge density, and GPS watermarking; ReportLab for 2-page Tax Invoices.

3. AI POD Verification Pipeline & Retake Logic

• **Stamp Detection:** Samples blue/violet rubber stamp ink saturation in receiving godowns. Ratio >= 0.001 confirms seal presence.
• **Signature Detection:** Evaluates edge density gradients in bottom 35% of LR receipt. Stroke density > 3.0 confirms receiver signature.
• **LR Number Match:** Validates OCR text against expected trip LR.
• **Shortage/Damage NLP:** Multilingual keyword scanner checks for 'short by 6 cartons', '■■ ■■■■', '■■■'. Discrepancies immediately route to NEEDS_REVIEW.
• **Retake Capping:** Drivers get at most 2 retakes. Subsequent defects route to human review queue with image as source of truth.

4. Billing Queue & Tally Prime Export

• **Review Queue:** Split-screen comparison with interactive document zoom on left and AI overrides on right.
• **Batch Invoicing:** Groups verified trips by Client and cycle, producing consolidated invoices.
• **Tally XML Export:** Exports valid Tally Sales Vouchers XML with Freight Income, CGST 6%, SGST 6%, and LR UDFs.
• **Tax Invoice PDF:** 2-page formal Tax Invoice with client details on Page 1 and certified watermarked POD on Page 2.

5. Scope Boundaries & Roadmap

• **Cut Scope (Out of Scope):** Live GPS tracking, trip route planning, driver fuel accounting.
• **Future Roadmap:** WhatsApp Cloud API bot for hired drivers, live Tally ODBC Windows daemon, edge TensorRT on rugged tablets.

6. Verification & Test Suite

All 9 automated unit and integration tests passing in 4.92s: AI verified pass, retake on missing stamp, retake capping at 2, shortage remark routing, image compression <300 KB, Tally XML schema, Tax Invoice PDF with POD, API trips list, and Dashboard economics math.